

CATERPILLAR® MACHINE DIAGNOSTICS

INDUSTRIAL INSPECTION & SYSTEM CONFIGURATION AUDIT

Machine ID:	123	Date of Inspection:	2026-07-19
Machine Name:	123	Lead Engineer:	John Administrator
Manufacturer:	123	Model / Category:	123 / CAT Hydraulic Excavator
Diagnostic Status:	Warning (Score: 90%)	Engine Version:	v2.4-deterministic

System Hardware & Firmware Configuration Audit

Below is a side-by-side comparison of the active telemetry and local software modules against original factory reference parameters.

Parameter	Factory Reference	Current Machine	Audit Status
Firmware Version	123	123	MATCHED
PLC Version	123		MISMATCH (WARNING)
CPU Architecture	123	123	MATCHED
RAM (Memory)	123	123	MATCHED
Storage	123	123	MATCHED
Sensor Count	123	123	MATCHED
Communication Ports	USB, COM1, Ethernet	COM1, Ethernet, USB	MATCHED
Installed Modules	Analog Input, Digital IO	Analog Input, Digital IO	MATCHED
Operating Temperature	Under 70 C	45.0 C	MATCHED
Power Supply Status	Stable	Stable	MATCHED

Detected Mismatches & Anomalies

1. PLC Version Mismatch [WARNING]

PLC configuration version mismatch. Expected '123', found ''.

AI-Powered Diagnostics Analysis

Machine Safety & Health Assessment

DEGRADED / ACTION REQUIRED (Health: 90%): Mismatches or anomalous sensor readings detected. Operational risks present.

Root Cause Analysis

The diagnostic engine flagged the following configuration mismatches: - PLC Version: PLC configuration version mismatch. Expected '123', found ''. Typical root causes include: field component replacement with unauthorized modules, out-of-date PLC programming flashes, firmware regression during maintenance cycles, or thermal cooling degradation.

Suggested Maintenance Worksteps

1. Connect Service Tool (CAT ET or Siemens TIA Portal) and flash target firmware and PLC image profiles matching reference configurations.

Crucial Safety Notes

- Verify power source disconnect before opening PLC enclosures.
- Ensure constant power backup is connected to the PLC logic controllers during the programming and flashing sequence.

Verification & Troubleshooting Steps

1. Run firmware CRC checksum validation command. Check serial console output for boot loader warnings.

PREPARED BY:

Automated Diagnostic Engine
CAT Diagnostic Systems

REVIEWED & SIGNED BY:

Lead Engineer: John Administrator
CAT Certified Field Inspector